

Swapnil Chhibber

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Education

Thapar Institute of Engineering and Technology

Patiala, India

B.E. in Computer Engineering | Cumulative GPA: 9.37/10.0

Sep. 2022 – Jun. 2026 (Expected)

Police DAV Public School, CBSE Board

Jalandhar, India

Class XII: 94.4% | Class X: 98.8% (2nd in District)

Relevant Coursework

Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Machine Learning, Software Engineering, Web Development, Probability and Statistics

Projects

Capstone Portal Management System | *Next.js, React 18, Material UI, Docker*

Status: Live, Ongoing Updates

- Built role-based student & mentor portals with **Next.js 14**, **React 18**, MUI, and TanStack Query, serving **3000+ students & 500+ faculty** with real-time dashboards.
- Integrated **JWT-auth**, role-based access, multi-stage forms, and client-side validation via Django REST APIs.
- Enabled **Dockerized** deployment with multi-env support, Blue-Green releases, and CI-ready infra ensuring **99.5% uptime**.

Project Management Platform | *JavaScript, PHP, Chart.js, WebSocket, GitHub API*

- Architected **15+ components** with vanilla JS and Chart.js, visualizing GitHub stats via **8+ chart types** for 200+ users.
- Automated repo management using **GitHub API**, webhooks, and contributor workflows, cutting manual effort by **90%**.
- Implemented **WebSocket-based chat** using ReactPHP + MySQL for project channels with sub-second latency.

MyChatApp – Real-Time Chat Platform | *MERN, Socket.IO, Zustand, JWT*

- Developed MERN-based chat app with **Socket.IO**, **Zustand** state management, and REST APIs.
- Achieved **100ms latency** 1:1 chats for 50+ users with real-time sync & online status.
- Secured auth with **JWT (access/refresh)**, bcrypt (10 rounds), HTTP-only cookies, and middleware-protected routes.
- Integrated **Cloudinary CDN** for 50MB profile uploads with image transformation; styled using **Tailwind + DaisyUI**.

MyShell – Custom Windows Shell Interface | *C++17, Windows API, CMake*

- Engineered a fully functional **Windows shell** with 8+ built-in commands, intelligent command parsing (quotes, pipes, redirection), and external process execution using **CreateProcess & ShellExecute**.
- Designed modular architecture with **RAII**, UTF-8 support, enum-based console theming, and **CMake-based multi-compiler builds** (GCC/MinGW, MSVC) with shell32.lib linking.

Experience

Creative Computing Society | *Executive Board Advisor*

Nov. 2022 – Present

- Lead a team of 200+ students across web development and marketing domains for technical society initiatives.
- Coordinated 5+ major events including HackTU (Official MLH hackathon, 2500+ participants), competitive programming contests, and technical workshops.

Technical Skills

Languages & Scripting: C++17, JavaScript, Python, SQL, PHP

Web Development: React, Next.js, HTML5, CSS3, Tailwind, Material-UI, Node.js, Express.js, Django REST

Databases & APIs: MongoDB, MySQL, REST APIs, JWT, WebSockets

Cloud & DevOps: GCP, AWS, Cloudinary, CI/CD, Blue-Green Deployment, Docker

Tools & Libraries: Git, VS Code, Postman, CMake, Zustand, TanStack Query, Chart.js, bcrypt, nanoid

Practices: Agile, Project Management, Team Collaboration, RAII Principles

Achievements

Awarded **INR 6.5 Lakhs** academic scholarship for outstanding academic performance.

Secured **1st place** in CCS Intra-Society Competitive Programming Contest among 50+ participants; won INR 5,000.

Solved **350+ LeetCode problems** and achieved 7+ badges across data structures and algorithms topics.

Earned **Google Cloud Skills Badge** for completing Introduction to Generative AI certification.